Duke University **October 15, 2020**
Math 105L: Laboratory Calculus and Functions I

Big Assignment 2

- This assignment covers work from Day 4-3 to Day 9-2.
- You have eight days to complete the assignment. It is due Oct 23 at 8am.
- You may use the worksheets, labs, and your notes, as well as any other resources.
- However, you may not work with anyone else on this assignment – it must be entirely your own work.
- Throughout the assignment, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers. For example, copying graphs from graphing software is insufficient 'work'.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.
- You must hand in a scanned version of this assignment on Gradescope and no other pages. If you need to do rough work, do it elsewhere before writing your answers.
- You must sign the statement at the bottom of this page.

Honor Code:

I have adhered to the Duke Community Standard and all the rules above in completing this assignment.

Signature: _____

Question 1 (16 points) Find the derivatives of the following functions. You may use any of the differentiation rules we learned.

(a) $f(x) = 6x^3 + 3x^2 + 4x$

(b) $g(x) = 2e^x + x^{2e} + \pi^e + 5^x$

(c) $h(x) = \frac{x^2+1}{x^3+2}$

(d) $i(x) = \ln(x)e^{x^2}$

Question 2 (12 points) In the Differentiating Power Functions and Polynomials lab, we showed the power rule (fill in the blank):

$$\frac{d}{dx}x^n = _____$$

We really only proved this for positive whole numbers n. In this question, we will use this power rule and the chain rule to find the derivative of $x^{-n} = \frac{1}{x^n}$.

(a) First, use the definition of the derivative to show that $\frac{d}{dx}\frac{1}{x} = -\frac{1}{x^2}$. You may use either definition of the derivative:

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \text{ or } f'(x) = \lim_{z\to x}\frac{f(z)-f(x)}{z-x}.$$

(b) Next, state the chain rule:

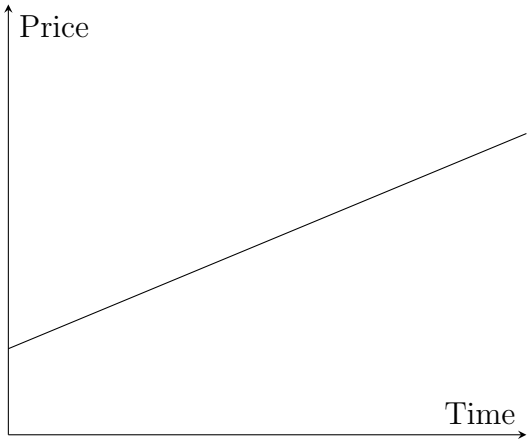
$$\frac{d}{dx}f(g(x)) = _____$$

(c) Lastly, use the fact that $x^{-n} = (x^n)^{-1}$, the chain rule, and the derivative you found in part (a) to show that $\frac{d}{dx}x^{-n} = -nx^{-n-1}$.

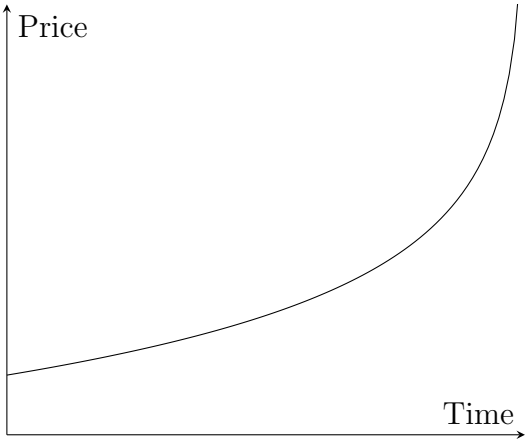
Question 3 (12 points) Suppose that a rate strategy for stock buying is one that buys the stock as soon as the rate of change of the stock price is positive. Suppose that an acceleration strategy buys as soon as the acceleration of the stock price is positive.

(a) For each of the following graphs of stock, label where each strategy buys the stock by labeling that point A or R respectively.

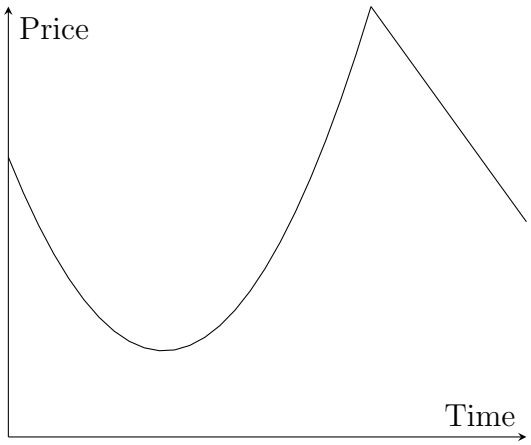
(b) If the stock is purchased, it is sold at the end of the period shown. Write under each graph whether the acceleration strategy made more money than the rate strategy (accel > rate), the rate strategy made more than the acceleration (accel < rate), or they made the same amount of money (accel = rate).



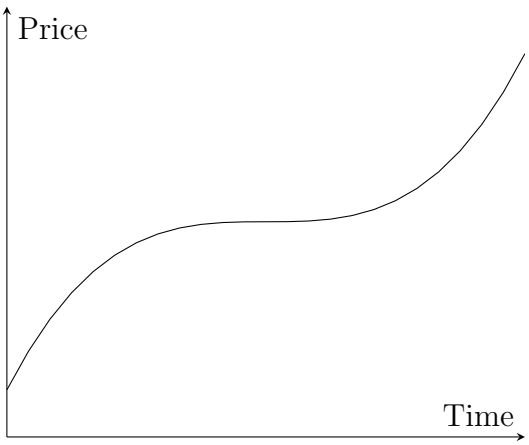
accel _____ rate



accel _____ rate



accel _____ rate



accel _____ rate

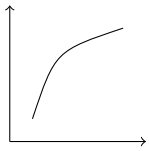
Question 4 (11 points) Consider the curve defined by the equation $xe^{-y} - 4y^2 = x^3$.

(a) Show that the point $(1, 0)$ is on the curve.

(b) Show that $\frac{dy}{dx} = \frac{e^{-y}-3x^2}{xe^{-y}+8y}$.

(c) Find the equation of the tangent line to the curve at the point $(1, 0)$.

Question 5 (10 points) Consider the following graph of a function $f(x)$, shown over the domain $1 \leq x \leq 5$:



Suppose that the tangent line to this curve at $x = 2$ is given by $y = 3x - 2$.

(a) Can you find a formula for $f(x)$? If so, do. If not, very briefly explain why not.

(b) What is $f^{-1}(4)$? Very briefly explain your answer.

(c) What is $f'(2)$? Very briefly explain your answer.

(d) Can you find $f''(2)$? If you can, do. If not, what can you say about it?

Question 6 (12 points) The following statements are false. Give a counterexample or counterargument in each case.

(a) If a function is not differentiable at a point $x = a$ in its domain, then the graph of the function has a sharp corner there.

(b) If a function $f(x)$ is invertible, then it is always increasing.

(c) Suppose we have 10 moles of a radioactive element whose decay is exponential, and an hour later, we have 5 moles. Then two hours from the start, we will have nothing left.

(d) If $f(x)$ and $g(x)$ are functions, then $f(g(x)) = g(f(x))$.

Question 7 (15 points) Solve the following equations, or show that there is no solution. Leave your answers in terms of e and/or logs.

(a) $4e^{2x-3} - 5 = e$

(b) $4e^{2x-3} + 5 = e$

(c) $4 \cdot 3^x = 7 \cdot 5^x$

Question 8 (12 points) This question concerns the exponential function $f(x) = a^x$.

(a) Briefly state why we cannot use the power rule to take the derivative of this function.

(b) Use the definition of the derivative:

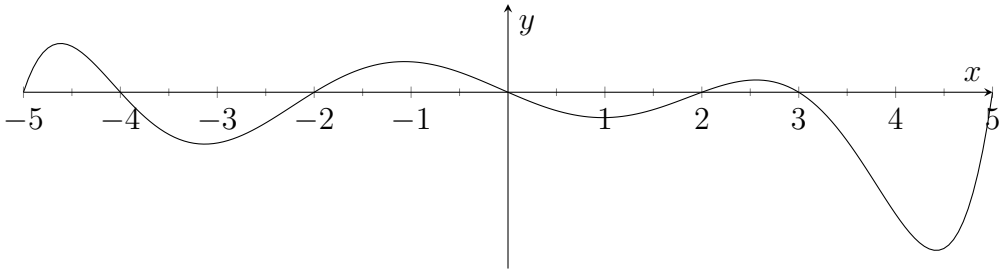
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

to show that $f'(x) = kf(x)$ for a constant k, and write down an expression for k.

(c) Assuming that $\frac{d}{dx}e^x = e^x$, use the chain rule to show that $k = \ln a$.

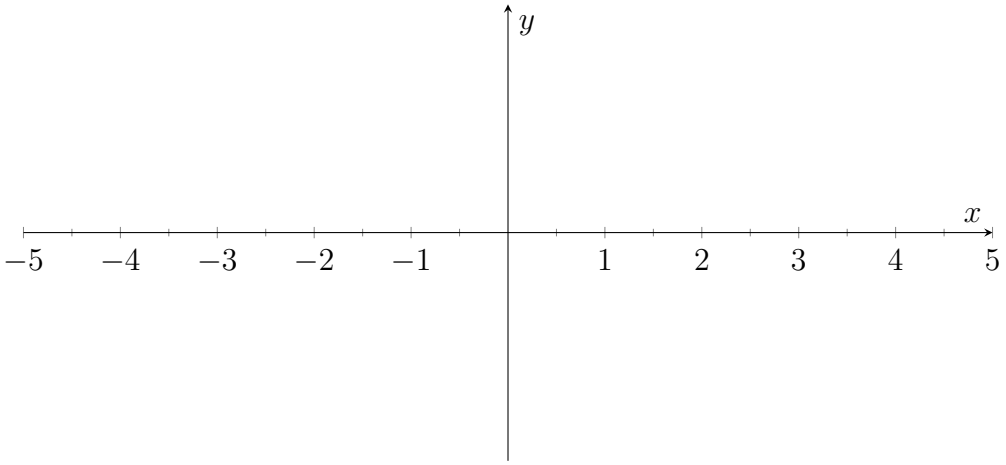
(Hint: $a^x = e^{\ln a^x} = e^{x \ln a}$.)

Question 9 (8 points) The following is a plot of the derivative of a function $f(x)$ on the domain $[-5, 5]$. This entire question concerns $f(x)$ and its derivatives.



Note: This is a plot of $f'(x)$, not $f(x)$!

(a) Sketch $f''(x)$ on the axes below:



(b) Find the interval(s) on which $f(x)$ is:

(i) increasing

(ii) concave down

Question 10 (12 points) Suppose that $R(t)$ is the percentage of religious people in a country t years since 1980. Suppose that $H(x)$ is the percentage of people in the country who wear hats when the percentage of religious people is $x\%$.

(a) Describe the function $f(t) = H(R(t))$.

(b) Explain what each of the following mean in a short sentence. Include units.

(i) $R(40) = 60$

(ii) $R'(40) = -1$

(iii) $H(60) = 20$

(iv) $H'(60) = 2$

(c) Given the above data, compute the rate at which the percentage of people wearing hats is changing in the country in 2020. Include units.

Bonus Question (2 points) Name the Democratic and Republican candidates for NC Lt. Governor, and something interesting about each of them.

Bonus Question Early Halloween Edition (2 points) Tell me a scary story, or draw a terrifying picture!